

CS 486/686: Assignment 2 (100 Points)

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Due Date: Thursday, July 9, 2026 at 11:59 pm (ET; Waterloo Time)

Instructions

- Submit solutions as two files to the Assignment 2 Dropbox on LEARN:
 - A file named `writeup.pdf` that contains all your written solutions;
 - A file named `code.py` that contains your code.

If your submission does not exactly contain two files named `writeup.pdf` and `code.py`, we will deduct 5 points from your final assignment mark.

- Use Python 3.12 or newer with NumPy installed to implement and run the code.
- No late assignment will be accepted.
- This assignment is to be done individually.

1 Question 1 [40pts]: Bayesian Networks and Variable Elimination

Suppose you are working for a financial institution and you are asked to implement a fraud detection system. You plan to use the following information:

- When the card holder is travelling abroad, fraudulent transactions are more likely since tourists are prime targets for thieves. More precisely, 1% of transactions are fraudulent when the card holder is travelling, where as only 0.4% of the transactions are fraudulent when they are not travelling. On average, 5% of all transactions happen while the card holder is travelling. If a transaction is fraudulent, then the likelihood of a foreign purchase increases, unless the card holder happens to be travelling. More precisely, when the card holder is not travelling, 10% of the fraudulent transactions are foreign purchases where as only 1% of the legitimate transactions are foreign purchases. On the other hand, when the card holder is travelling, then 90% of the transactions are foreign purchases regardless of the legitimacy of the transactions.
- Purchases made over the internet are more likely to be fraudulent. This is especially true for card holders who don't own any computer. Currently, 60% of the population owns a computer and for those card holders, 1% of their legitimate transactions are done over the internet, however this percentage increases to 2% for fraudulent transactions. For those who don't own any computer, a mere 0.1% of their legitimate transactions is done over the internet, but that number increases to 1.1% for fraudulent transactions. Unfortunately, the credit card company doesn't know whether a card holder owns a computer, however it can usually guess by verifying whether any of the recent transactions involve the purchase of computer related accessories. In any given week, 10% of those who own a computer purchase with their credit card at least one computer related item as opposed to just 0.1% of those who don't own any computer.

- a [10 pts] Construct a Bayes Network that your fraud detection system can use to identify fraudulent transactions.

What to hand in: Show the graph defining the network and the Conditional Probability Tables associated with each node in the graph. This network should encode the information stated above. Your network should contain exactly six nodes, corresponding to the following binary random variables:

- *OC* – card holder owns a computer.
- *Fraud* – current transaction is fraudulent.
- *Trav* – card holder is currently travelling.
- *FP* – current transaction is a foreign purchase.
- *IP* – current purchase is an internet purchase.

- *CRP* – a computer related purchase was made in the past week.

The arcs defining your Bayes Network should accurately capture the probabilistic dependencies between these variables.

- b [25 pts] What is the prior probability (i.e., before we search for previous computer related purchases and before we verify whether it is a foreign and/or an internet purchase) that the current transaction is a fraud?

What is the probability that the current transaction is a fraud once we have verified the following 3 facts: (1) it is a foreign transaction; (2) it is not an internet purchase; and (3) the card holder purchased computer related accessories in the past week? (Note: we are describing one query, not three separate queries.)

What to hand in: Indicate the two queries (i.e., $Pr(variables|evidence)$) you used to compute those probabilities. Show the factors computed at each step of variable elimination. For each step, show a formula giving the new factor to be created, e.g.

$$f_7(B) = \sum_A [f_2(B, A) \cdot f_3(A) \cdot f_6(A, B)]$$

B	$f_7(B)$
t	0.007
f	0.640

And show the resulting factor (you don't need to show all steps of the calculation). Use the following variable ordering when summing out variables in variable elimination: *Trav, FP, Fraud, IP, OC, CRP*. No marks are awarded for answering the questions using external software.

- c [5 pts] Suppose you are not a very honest employee and you just stole a credit card. You know that the fraud detection system uses the Bayes net designed earlier but you still want to make an important purchase over the internet. What can you do prior to your internet purchase to reduce the risk that the transaction will be rejected as a possible fraud? This question is independent of question 1b.

What to hand in: State the action taken and indicate by how much the probability of a fraud gets reduced. Follow the same instructions as for Question 1b to show that the probability decreases. You don't have to replicate all the calculations, only those that change.

2 Question 2 [60pts]: Markov Decision Processes

In this question, we will continue to study the grid world discussed in class. We will vary the reward function $R(s)$ for any state $s \neq s_{24}$ and $s \neq s_{34}$ and investigate how the optimal policy changes as $R(s)$ changes.

Assume that $R(s_{24}) = -1$, $R(s_{34}) = +1$, and the discount factor $\gamma = 1$. The transition model stays the same: the agent moves in the intended direction with probability 0.8, moves to the left of the intended direction with probability 0.1, and moves to the right of the intended direction with probability 0.1.

	1	2	3	4
1				
2		X		-1
3				+1

- a [20 pts] Consider $R(s) = -1.0, \forall s \neq s_{24} \wedge s \neq s_{34}$. In iteration 0, let the true utility of every state to be 0. Implement the value iteration algorithm. Execute the value iteration algorithm until the true utilities of all the states have converged. Each true utility has converged when it stops changing up to three decimal places.

Please show the optimal policy in a table format. See the table below for an example. s_{23} shows a case when there is a tie between multiple actions for the optimal policy.

	1	2	3	4
1	↑	↓	←	→
2	↑	X	↓↑	-1
3	←	→	↑	+1

What to hand in:

- Show the optimal policy of the MDP. If there is a tie between multiple actions, show all of them as the optimal policy for a state.
 - Show the true utilities of all the states until the iteration when all the true utilities have stopped changing up to three decimal places. That is, execute the algorithm until the earliest iteration K such that the true utilities for all the states are the same in iteration $K - 1$ and iteration K .
 - A program you used to execute the value iteration algorithm and produce the output in the file *code.py*.
- b [20 pts] Determine how the optimal policy changes when $R(s)$ changes from -1.6 to -0.5 . You only need to consider the values of $R(s)$ up to one decimal place. That is, you only need to consider $R(s) = -1.6, -1.5, -1.4, -1.3, -1.2, -1.1, -1.0, -0.9, -0.8, -0.7, -0.6, -0.5$.

What to hand in:

- Divide up the values of $R(s)$ into ranges such that the optimal policy for each range of values remains the same.

- State each range of values and state the optimal policy. If there is a tie between multiple actions, show all of them as the optimal policy for a state.
- Indicate how the optimal policy changes from the last value in one range to the first value in the next range.

c [20 pts] Determine how the optimal policy changes when $R(s)$ changes from -0.08 to -0.03 .

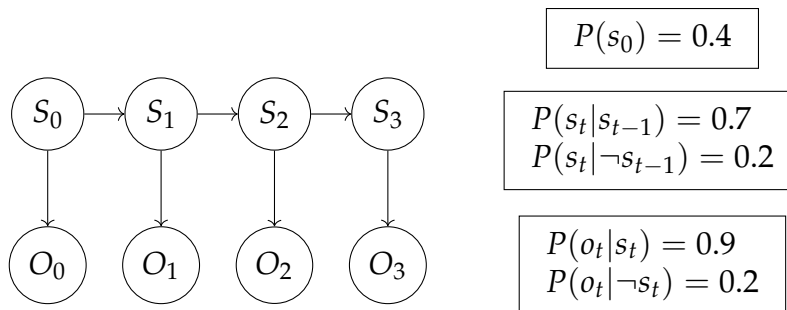
You only need to consider the values of $R(s)$ up to two decimal places. That is, you only need to consider $R(s) = -0.08, -0.07, -0.06, -0.05, -0.04, -0.03$.

What to hand in:

- Divide up the values of $R(s)$ into ranges such that the optimal policy for each range of values remains the same.
- State each range of values and state the optimal policy. If there is a tie between multiple actions, show all of them as the optimal policy for a state.
- Indicate how the optimal policy changes from the last value in one range to the first value in the next range.

3 Bonus Question (Optional) [20 extra pts]: Viterbi Decoding

Consider a hidden Markov model (HMM) with four time steps illustrated below:



Given observations of $O_0 = \text{True}$, $O_1 = \text{True}$, $O_2 = \text{True}$, $O_3 = \text{False}$, determine the sequence of hidden states $S_0^*, S_1^*, S_2^*, S_3^*$ that maximizes the posterior probability:

$$S_0^*, S_1^*, S_2^*, S_3^* = \arg \max_{S_0:S_3} P(S_0, S_1, S_2, S_3 | O_0, O_1, O_2, O_3).$$

You may solve this either by hand or by writing a program.

What to hand in:

- The optimal sequence of hidden states $(S_0^*, S_1^*, S_2^*, S_3^*)$.
- If solving by hand, clearly show your step-by-step Viterbi decoding procedure (including intermediate probabilities and back-pointers).
- If solving by program, submit your code in the file *code.py*, along with the resulting optimal hidden state sequence.